

Zero-Trust Model & Default-Deny Enforcement

Zero-Trust Model & Default-Deny Enforcement

Revision: 1 Last modified: 2026-06-25T12:00:00Z

Master technical specification — Volume 5 (Security & Privacy), nano-detail document **zero-trust-and-default-deny**. Deepens the **S1 zero-trust default-deny** invariant of [04-security-privacy-pki.md §0.1/§1.3] into an implementation-grade contract: the no-implicit-trust-by-network-location model, the two policy enforcement points (control-plane compile-time + data-plane runtime), the least-privilege two-artifact output (AllowedIPs + verdict map), the need-to-know peer-filtering algorithm, fail-closed semantics on control-plane/policy unavailability, and a fully worked policy evaluation with Mermaid. SPEC ONLY — describe the implementation, do not build the product. This document **references** the policy compiler ([v03-control-plane/svc-policy.md]) and the data-plane verdict-map renderer ([v02-data-plane/routing-and-addressing.md]) rather than redefining them; it owns the *zero-trust property* and the *default-deny enforcement contract* that bind both. Sources cited inline by id: [04_ARCH §N], [04_P1 §N], [research-pki_pq_nat], [SYNTHESIS §N], and sibling specs by filename. Invariant ids: S1–S11 ([04-security-privacy-pki.md §0.1]); P1–P8 ([v03-control-plane/svc-policy.md §0.1]); I5/I6/R1–R4 ([v02-data-plane/routing-and-addressing.md §0.2]). Unproven facts are flagged **UNVERIFIED** per constitution §11.4.6 — never fabricated.

Table of contents

- 0. Position, ownership & governing invariants
- 1. The zero-trust model — no implicit trust by network location
- 2. Authenticate AND authorize — the two gates every peer crosses
- 3. The two policy enforcement points (compile-time + runtime)
- 4. Default-deny enforcement — the empty policy denies all
- 5. Least-privilege artifacts — AllowedIPs + verdict map
- 6. Need-to-know peer filtering — the algorithm
- 7. Fail-closed semantics — deny on unavailability, never allow
- 8. Worked policy evaluation (end to end)
- 9. Edge cases (defined behavior, no guessing)

- 10. Test & validation mapping (§11.4.169)
 - Sources verified
-

0. Position, ownership & governing invariants

Zero-trust is the spine of HelixVPN: routing, no-logging, and revocation are all *properties of the build* only because nothing is trusted by default [04_ARCH §7]. The mechanisms that *realize* zero-trust live in other documents — the policy compiler in [v03-control-plane/svc-policy.md], the verdict-map renderer in [v02-data-plane/routing-and-addressing.md], the enrollment/PKI/mTLS in [04-security-privacy-pki.md]. **This document owns the zero-trust contract** that binds them: what "no implicit trust" means precisely, where it is enforced, and what "default-deny" and "fail-closed" must do under every condition.

This document **owns**: the zero-trust property statement (§1), the authenticate-AND-authorize gate model (§2), the two-enforcement-point map (§3), the default-deny floor (§4), the least-privilege artifact contract (§5), the need-to-know filtering algorithm restated as a zero-trust obligation (§6), and the fail-closed semantics (§7).

It does **not** own: the ACL DSL grammar, the compile algorithm, the version lifecycle, or the protobuf wire — all [v03-control-plane/svc-policy.md]; the FIB/4via6 mechanics or the nftables/eBPF rendering — [v02-data-plane/routing-and-addressing.md]; the auth handshakes themselves — [04-security-privacy-pki.md].

0.1 Governing invariants (every clause obeys these)

#	Invariant	Owner
S1	Zero-trust default-deny — no peer reaches anything without an explicit compiled policy rule; the empty policy denies all.	[04-security-privacy-pki.md §0.1]
S3	Need-to-know map distribution — a device's map contains only the peers/routes its policy grants; filtered server-side before the wire.	[04-security-privacy-pki.md §0.1]
S4	Two-channel auth — control (mTLS leaf cert) and data (WG Noise IK) are separate and both required.	[04-security-privacy-pki.md §0.1]
P1	Default-deny, need-to-know — a device appears in another's map only via a compiled accept edge; no deny verb (absence is denial).	[v03-control-plane/svc-policy.md §0.1]
P4	Fail-closed — a compile error rejects the change; it never degrades to allow-all or silently keeps the prior policy.	[v03-control-plane/svc-policy.md §0.1]
I6	Default-deny at the data plane — no peer reaches anything without an explicit rule expressed as AllowedIPs + an edge verdict map.	[v02-data-plane/routing-and-addressing.md §0.2]

#	Invariant	Owner
R2	Peers delivered already policy-filtered (need-to-know): a node never learns of networks it cannot reach.	[v02-data-plane/routing-and-addressing.md §0.2]
R3	The verdict map fails closed — absent/stale compiled policy $\Rightarrow$ drop.	[v02-data-plane/routing-and-addressing.md §0.2]

1. The zero-trust model — no implicit trust by network location

1.1 The principle

Classic VPNs are **perimeter** models: once you are "on the network" (past the VPN concentrator) you are implicitly trusted to reach everything on it. HelixVPN rejects this. The overlay is **zero-trust**: being enrolled, online, and cryptographically connected grants a device **nothing** by itself. Network *location* — "I have an overlay address; I completed a WG handshake to the gateway" — confers **zero** reachability. Every reachable peer and every reachable CIDR is the result of an *explicit, compiled, per-device* policy grant [04_ARCH §7, S1, P1].

The contrast, stated as a contract:

Perimeter VPN (rejected)	HelixVPN zero-trust (this spec)
"On the network" $\Rightarrow$ reach the whole subnet	"On the overlay" $\Rightarrow$ reach nothing until policy grants it
Trust derives from IP/segment location	Trust derives from <i>identity</i> (cert) plus <i>authorization</i> (compiled rule)
One firewall at the perimeter	Per-device least-privilege grant compiled into two enforcement layers
A breached host can move laterally freely	A breached host can reach only its compiled grant (microsegmentation)
Map = the whole topology	Map = only the need-to-know slice (S3/R2)

1.2 What "no implicit trust" forbids — the closed list

A conforming implementation MUST NOT, anywhere, grant reachability from any of these implicit signals (each is the perimeter-trust anti-pattern):

1. **Enrollment** — a freshly-enrolled device with no policy grant has an **empty** AllowedIPs and **no** verdict-map entry; it is online but isolated ([04-security-privacy-pki.md §1.3]).
2. **Online presence** — being in Redis presence state grants nothing.
3. **Overlay-address adjacency** — sharing a tenant ULA /48 grants nothing; two devices in the same /48 cannot reach each other without a rule.

4. **A completed WG handshake** — the data channel being up grants nothing the map did not already authorize (the map *is* the set of peers it may handshake with).
5. **Group membership alone** — membership resolves to a device set, but a group reaches a destination only via an accept rule naming that group as src ([v03-control-plane/svc-policy.md §4.1]).

There is no "default LAN", no "trusted subnet", no implicit gateway full-tunnel.

Absence of a rule is denial (P1) — there is no deny verb in the MVP model because there is nothing to subtract from; the base state is already deny-all.

flowchart LR

```

    E["Device enrolled + online + WG up"]
    Q{"Is there a compiled<br/>accept rule granting it?"}
    E --> Q
    Q -- "NO (the default)" --> D["DENY – empty AllowedIPs, no verdict entry, not in any peer's map (S1/P1/I6)"]
    Q -- "YES" --> A["ALLOW – exactly the granted peer+CIDR+ports, nothing more (least privilege)"]
  
```

2. Authenticate AND authorize — the two gates every peer crosses

Zero-trust requires **both** authentication (*who are you*) and authorization (*what may you reach*). HelixVPN keeps these on two separate, never-conflated channels (S4) — a property the threat model relies on (compromising one does not yield the other, [threat-model.md §5.2/§5.4]).

Gate	Question	Mechanism	Owner	Failure mode
G1 — Authenticate (control)	<i>Who are you?</i>	mTLS with a ≤ 24 h CA-signed device leaf cert on every Coordinator RPC	[04-security-privacy-pki.md §1.2/§4]	no valid cert $\Rightarrow$ stream refused; nothing learned
G2 — Authorize (compile)	<i>What may you reach?</i>	the device's compiled CompiledPolicy slice (VisibleTo/AllowedIPs/Verdicts)	[v03-control-plane/svc-policy.md §3]	empty slice $\Rightarrow$ reach nothing (default-deny)
G3 — Authenticate (data)	<i>Are you the keyholder?</i>	WG Noise IK handshake to a peer pubkey delivered by G2's filtered map	doc 01 / [04-security-privacy-pki.md §1.2]	wrong key $\Rightarrow$ handshake fails
G4 — Enforce (runtime)	<i>Is this packet</i>	AllowedIPs cryptokey routing	[v02-data-plane/routing-and-	outside grant $\Rightarrow$ dropped

Gate	Question	Mechanism	Owner	Failure mode
	<i>within the grant?</i>	+ verdict map at the edge	addressing.md §6]	

A device must pass **G1 to learn anything, G2 to be granted anything, G3 to connect to a peer, and G4 to send a packet to it.** The four gates are independent; bypassing one does not bypass the others. In particular, even a device that somehow obtained a peer's pubkey out-of-band cannot reach it: G4's AllowedIPs + verdict map drop the packet unless G2 compiled the grant.

sequenceDiagram

```

    autonumber
    participant Dev as Device (helix-core)
    participant Coord as coordinator
    participant Pol as policy (CompiledPolicy)
    participant Edge as Rust edge (verdict map)
    Dev->>Coord: WatchNetworkMap (mTLS leaf cert)    %% G1
    authenticate
    Coord->>Coord: bind stream to device_id from the CERT (never
    client-supplied)
    Coord->>Pol: read CompiledPolicy.VisibleTo[device_id]    %% G2
    authorize (need-to-know)
    Pol-->>Coord: only the granted peers + AllowedIPs + Verdicts
    Coord-->>Dev: NetworkMap (filtered slice ONLY) (S3/R2)
    Dev->>Edge: WG Noise IK to a granted peer's pubkey    %% G3
    authenticate (data)
    Dev->>Edge: packet to granted CIDR:port
    Edge->>Edge: AllowedIPs match? verdict ACCEPT?    %% G4 enforce
    (runtime)
    Edge-->>Dev: forward iff within grant; else DROP (default-
    deny, I6/R3)

```

3. The two policy enforcement points (compile-time + runtime)

Zero-trust in HelixVPN is enforced at **two** points — once when intent becomes least-privilege artifacts (control plane, compile-time) and once when packets meet those artifacts (data plane, runtime). This is the PDP/PEP split (policy decision point / policy enforcement point) made concrete.

Enforcement point	Where	What it decides/enforces	Artifact	Owner
EP1 — Compile-time (PDP)	Go control plane	resolves the ACL into per-device need-to-know reachability;	CompiledPolicy (§5)	[v03-control-plane/svc-policy.md §4]

Enforcement point	Where	What it decides/enforces	Artifact	Owner
EP2a — Distribution filter (PEP at the wire)	coordinator	rejects anything default-deny does not grant filters each device's map to its VisibleTo slice <i>before</i> serialization (S3/R2)	filtered NetworkMap	coordinator (doc 03)
EP2b — Runtime (PEP at the packet)	Rust edge / device	drops any packet outside AllowedIPs (cryptokey routing) and outside the verdict map (default DROP)	kernel WG AllowedIPs + nftables/eBPF verdict map	[v02-data-plane/routing-and-addressing.md §6]

The two-layer runtime (EP2b) is deliberate defense-in-depth: WG's own AllowedIPs cryptokey routing is a *coarse* CIDR gate, and the verdict map is the *fine* port-level gate; **both must allow** for a packet to pass, and **both default to deny** ([v03-control-plane/svc-policy.md §3 P3], I6). A bug in one layer is caught by the other.

flowchart TB

```

    subgraph PDP["EP1 – Policy Decision (compile-time, control plane)"]
        Spec["ACL spec (intent)"] --> Comp["compile() pure + deterministic (P2)"]
        Comp --> CP["CompiledPolicy {VisibleTo, AllowedIPs, Verdicts, Via6, ExitNodes}"]
    end
    subgraph PEPw["EP2a – PEP at the wire (coordinator)"]
        CP --> Filt["filter to VisibleTo[device] (S3/R2)"]
        Filt --> Map["NetworkMap (need-to-know slice)"]
    end
    subgraph PEPp["EP2b – PEP at the packet (edge/device, runtime)"]
        Map --> AIP["WG AllowedIPs (coarse CIDR gate)"]
        Map --> VM["verdict map (fine port gate, default DROP)"]
        AIP --> Out["packet forwarded IFF both ALLOW; else DROP (I6/R3)"]
        VM --> Out
    end

```

4. Default-deny enforcement — the empty policy denies all

4.1 The floor: the empty policy

The base case proves the model. A tenant with **no ACL rules** (only groups/hosts, or nothing at all) compiles to an **all-empty** CompiledPolicy: every map is empty, every AllowedIPs is empty, every verdict entry is absent. Result: **everyone is denied everything** ([v03-control-plane/svc-policy.md §10 E1]). This is **valid, not an error** — it is the default-deny floor (P1/S1). Reachability is *added* by rules; it is never the starting state.

4.2 Default-deny at each layer (the closed table)

Layer	"No grant" state	Effect
CompiledPolicy.VisibleTo[d]	empty set	device d learns of no peers (S3)
CompiledPolicy.AllowedIPs[d][peer]	absent	WG cryptokey routing drops all packets to/from peer (I6)
CompiledPolicy.Verdicts[d][peer]	absent / default	edge verdict map default verdict is DROP (R3)
coordinator distribution	peer not in VisibleTo	peer never serialized into d's map (S3/R2)
edge, no compiled policy at all	policy: None	the RouteEngine fails closed — every packet dropped ([v02-data-plane/routing-and-addressing.md §1 policy: Option<CompiledPolicy> None ⇒ fail-closed (R3)])

4.3 Least-privilege is the only addition path

A grant adds **exactly** the (peer, CIDR, ports) the rule names and **nothing more** ([v03-control-plane/svc-policy.md §4.1]):

- A `host:554,80` grant adds AllowedIPs = [host CIDR] *and* a Verdicts entry restricting to TCP/UDP 554 and 80 — not all ports.
- A `*:*` grant is the operator's explicit choice to grant broad reach (no implicit narrowing, E2); it is still *compiled per device*, not an ambient "trust the network" rule.
- A device absent from ExitNodes cannot full-tunnel through the gateway, even if it can reach individual CIDRs ([v03-control-plane/svc-policy.md §4.3]).

No rule ever *widens* beyond its tokens, and the compiler is **pure + deterministic** (P2) so the same intent always yields the same least-privilege artifact — a property the §10 tests assert byte-for-byte.

5. Least-privilege artifacts — AllowedIPs + verdict map

The compiler emits the single source of zero-trust truth: `CompiledPolicy`. Restated from [v03-control-plane/svc-policy.md §3] for the zero-trust contract (this document references it, does not redefine the Go types):

```
// from internal/policy/compiled.go – the per-(tenant,version)
// zero-trust artifact.
type CompiledPolicy struct {
    Version    Version
    SpecHash   [32]byte // sha256(canonicalJSON(spec)) – ties
              artifact to intent (audit + dedup)

    // ARTIFACT 1 – need-to-know visibility (S3/P1): src device ->
    // reachable peer set.
    VisibleTo  map[DeviceID]map[DeviceID]struct{}
    // ARTIFACT 1b – coarse WG AllowedIPs (CIDR-only, I6): src ->
    // peer -> CIDRs.
    AllowedIPs map[DeviceID]map[DeviceID][]netip.Prefix
    // ARTIFACT 2 – fine port-level verdict map (R3 default-DROP):
    // src -> peer -> port rules.
    Verdicts   map[DeviceID]map[DeviceID][]PortRule
    // 4via6 routes for colliding IPv4 LANs (D4): src ->
    // peer(connector) -> mappings.
    Via6       map[DeviceID]map[DeviceID][]Via6Route
    // exit-node grants: devices permitted to full-tunnel through
    // the gateway.
    ExitNodes  map[DeviceID]struct{}
}
```

The three reachability fields are the zero-trust enforcement surface:

Field	Role	Default-deny meaning	Enforced at
VisibleTo[d]	who d may <i>learn about</i>	empty $\Rightarrow$ learns nothing (S3)	EP2a coordinator filter
AllowedIPs[d][peer]	which CIDRs d may <i>reach</i> via peer	absent $\Rightarrow$ all dropped (I6, coarse)	EP2b WG cryptokey routing
Verdicts[d][peer]	which ports of those CIDRs	absent $\Rightarrow$ default DROP (R3, fine)	EP2b nftables/eBPF verdict map

SpecHash ties every artifact back to the exact authored intent (a zero-trust audit anchor — you can always prove *which* spec produced *this* grant), and Version makes rollback a flip to a previously-compiled artifact, never a recompile under pressure ([v03-control-plane/svc-policy.md §6 P5]).

The Peer protobuf the coordinator streams projects 1:1 from these fields — allowed_ips from AllowedIPs[self][peer], verdicts from Verdicts[self][peer] ([v03-control-plane/svc-policy.md §7.2]). The wire delivers *only the filtered slice*, so a device physically never receives a peer it cannot reach (S3/R2) — least-privilege is enforced even at the byte level, not just at the firewall.

6. Need-to-know peer filtering — the algorithm

Need-to-know (S3/R2) is the zero-trust *distribution* obligation: filter the map **server-side, before the wire**, so a device's map contains only its grant. A device that should not reach conn-B never learns conn-B exists — preventing both reconnaissance and accidental over-share.

6.1 The filtering algorithm

For a device *d* opening WatchNetworkMap (authenticated as device_id = *d* by its mTLS cert, G1/§2), the coordinator builds *d*'s NetworkMap thus (deriving strictly from CompiledPolicy, [v03-control-plane/svc-policy.md §3/§4.1]):

```
buildMap(d, cp /* active CompiledPolicy */ -> NetworkMap:
  # ZERO-TRUST RULE: start from nothing; add only what cp grants.
  m := NetworkMap{ peers: [] } # default-deny base
  (S1/P1)
  for peer in sortedKeys(cp.VisibleTo[d]): # ONLY the need-to-
    know set (S3)
    if snapshot.Devices[peer].Revoked: continue # revoked
    never visible (P1)
    p := Peer{
      device_id: peer,
      wg_pubkey: snapshot.Devices[peer].WGPubKey, # so
      d can do WG IK (G3)
      allowed_ips: cp.AllowedIPs[d][peer], #
      coarse CIDR grant (I6)
      verdicts: cp.Verdicts[d][peer], #
      fine port grant (R3)
      via6: cp.Via6[d][peer], #
      4via6 for colliding LANs (D4)
      is_connector: snapshot.Devices[peer].Kind == CONNECTOR,
      endpoint: gatewayRelay, # MVP
      relays via gateway
    }
    m.peers.append(p)
  m.dns := tunnelDNS; m.exit_allowed := (d in cp.ExitNodes) #
  exit only if granted (§4.3)
```

```

return m                                     # contains
EXACTLY d's grant – nothing else

```

Three zero-trust properties the algorithm guarantees:

1. **Server-authoritative identity** — *d* is taken from the *cert*, never from a client field, so a device cannot ask for another device's map ([threat-model.md §5.5 T-COORD-S-1]).
2. **Filter before serialize** — peers outside *VisibleTo[d]* are never placed in the message, so the over-share threat is closed at the data structure, not at a later firewall (S3/R2).
3. **Revoked-exclusion** — a revoked peer never enters *VisibleTo(P1)* and is re-checked here (belt-and-suspenders), tying into the <1 s revocation pipeline (S5, [04-security-privacy-pki.md §4.6]).

6.2 Delta updates preserve need-to-know

When policy changes, the coordinator pushes a **MapDelta** (peer upsert/remove) to each affected open stream — the same need-to-know filter applies to the delta: a device only ever receives upserts for peers in its new *VisibleTo* and removals for peers leaving it ([v03-control-plane/svc-policy.md §11]). A device never learns a peer it lost reach to *was* reachable beyond the removal itself; it never learns a peer it never had.

7. Fail-closed semantics — deny on unavailability, never allow

Zero-trust must hold **especially** when something breaks. Every unavailability path resolves to **deny**, never to allow (P4/R3). "When in doubt, drop" is the contract.

7.1 The fail-closed table (closed enumeration)

Failure condition	Where	Behavior	Invariant
Policy spec fails to compile	control plane, EP1	reject the update; no row inserted; prior active policy stays live (never allow-all)	P4
dst CIDR not covered by an advertised prefix	EP1 dry-run	blocking ERR_HOST_NOT_ADVERTISED; change rejected	P4/P7
Edge has no compiled policy (policy: None)	EP2b edge	drop every packet (the Option<CompiledPolicy> is None)	R3
	EP2b edge		

Failure condition	Where	Behavior	Invariant
Compiled policy is stale (delta not yet applied)		fail-static: keep enforcing the <i>old</i> policy (old tunnels keep forwarding under old rules) — never fail-open to allow-all	R3 / [v03-control-plane/svc-policy.md §10.3]
Control plane unreachable from a device	device	device keeps its <i>last</i> filtered map (fail-static to its existing grant); cannot <i>widen</i> without a new push; kill-switch keeps the leak window closed (S9)	R3 / S9
Coordinator stream drops	coordinator	the device re-authenticates (G1) and re-fetches; until then it has only its prior grant	R2/R3
Device revoked mid-flight	control + edge	revocation removes the peer + closes the stream in <1 s (S5); a revoked grant is re-checked at activation (E5)	S5 / P4

7.2 Fail-static is not fail-open — the honest distinction (§11.4.6)

"Fail-closed" here has two faces, both deny-leaning, and the spec is precise about which:

- **Fail-closed (reject)** — a *new* change that cannot be safely compiled/validated is **rejected**; the system does not adopt it (P4). Default-deny is preserved by *not widening*.
- **Fail-static (hold)** — when a *delta* is in flight, edges keep enforcing the **last known good** policy until the new one lands; they never drop to allow-all and never blindly widen. This is the <1 s convergence floor: a sub-second window of *old-but-correct* enforcement, never a window of *open* enforcement ([v03-control-plane/svc-policy.md §10.3]).

Neither face ever resolves to "allow because we're unsure". The honest boundary: the system guarantees **no fail-open gap**; it does **not** claim globally-atomic instantaneous convergence — there is a measured sub-second window where different edges may enforce different (*both valid, both deny-correct*) versions. **UNVERIFIED:** the exact p99 of this window under 10k concurrent streams is a measured number the soak test produces ([v03-control-plane/svc-policy.md §10.3]) — this document states the *target* (<1 s), not a measured result.

stateDiagram-v2

```

    [*] --> DenyAll: no policy (None) – fail-closed floor (R3)
    DenyAll --> Enforcing: Activate(v) → edge receives compiled artifact
    Enforcing --> Enforcing: MapDelta applied (converged, <1s target)
    Enforcing --> HoldingOld: delta in flight (fail-static – keep OLD policy)
    HoldingOld --> Enforcing: delta lands
    Enforcing --> Rejected: new spec fails compile/dry-run (P4)

```

```

    Rejected --> Enforcing: prior active policy UNCHANGED (never
allow-all)
    Enforcing --> DenyPeer: device.revoked → peer removed <1s (S5)
    note right of HoldingOld
        Fail-STATIC = old-but-correct deny enforcement.
        NEVER fail-OPEN to allow-all. No leak window (S9).
    end note

```

8. Worked policy evaluation (end to end)

Reusing the running example of [v03-control-plane/svc-policy.md §2.1/§4.3] to show zero-trust

- default-deny operating concretely. Tenant devices: alice-laptop, bob-laptop (group admins), carol-laptop (group contractors), conn-A (serves 10.10.0.0/24, site 1), conn-B (serves 192.168.50.0/24, site 2), gateway gw.

Intent (the ACL):

```

{
  "acls": [
    { "action": "accept", "src": ["group:admins"],      "dst":
["*:*"] },
    { "action": "accept", "src": ["group:contractors"], "dst":
["warehouse-cams:554,80"] }
  ],
  "exitNodes": ["group:admins"] // only admins may full-tunnel
}

```

8.1 Evaluation — what each device may reach (and may NOT)

flowchart LR

```

subgraph admins["group:admins → *:~ (+ exit)"]
  AL[alice-laptop]
  BL[bob-laptop]
end
CL[carol-laptop]
subgraph conns["connectors"]
  CA["conn-A 10.10.0.0/24 site1"]
  CB["conn-B 192.168.50.0/24 site2"]
end
GW[(gateway exit)]

```

```

AL -->|"AllowedIPs 10.10.0.0/24 + via6 site1; all ports"| CA
AL -->|"AllowedIPs 192.168.50.0/24 + via6 site2; all ports"| CB
AL -->|"0.0.0.0/0 ::/0 EXIT (in ExitNodes)"| GW
BL --> CA
BL --> CB
BL --> GW
CL -->|"AllowedIPs 10.10.0.0/24 ; Verdicts TCP 554,80 ONLY ;
via6 site1"| CA

```

```
CL -. "DEFAULT-DENY: no rule grants office-lan → NOT in
VisibleTo, no AllowedIPs" .-> CB
CL -. "DEFAULT-DENY: carol ∉ ExitNodes → no full-tunnel" .-> GW
```

8.2 The precise compiled slice for carol-laptop (least privilege made concrete)

```
VisibleTo[carol-laptop]          = { conn-
A }                               # need-to-know: only conn-A (S3)
AllowedIPs[carol-laptop][conn-A] =
[ 10.10.0.0/24 ]                  # coarse grant (I6)
Verdicts[carol-laptop][conn-A]   = [ {10.10.0.0/24, ANY,
554,554}, # fine: ports 554 + 80 ONLY (R3)
{10.10.0.0/24, ANY, 80,80} ]
Via6[carol-laptop][conn-A]       = [ {10.10.0.0/24,
fd7a:helix:rrrr:0001::/96} ]    # 4via6 (D4)
ExitNodes                         ∄ carol-
laptop                           # no full-tunnel (§4.3)
# carol-laptop has NO entry for conn-B / office-lan → DEFAULT-DENY
(S1/P1/I6)
```

8.3 Runtime enforcement of carol's slice (EP2b, two layers)

A packet from carol-laptop:

Destination attempt	WG AllowedIPs (coarse)	Verdict map (fine)	Result
10.10.0.5:554 (TCP)	match 10.10.0.0/24 ✓	match 554 ✓	ALLOW — the granted camera port
10.10.0.5:22 (SSH)	match 10.10.0.0/24 ✓	no rule → default DROP	DENY — port not granted (least privilege)
192.168.50.5:any	no AllowedIPs → cryptokey-route DROP	(never reached)	DENY — office- lan not granted (S1)
full-tunnel 8.8.8.8	not in ExitNodes → no 0.0.0.0/0 grant	—	DENY — not an exit node

The two layers are visible: even where the *coarse* CIDR layer would pass (10.10.0.0/24), the *fine* verdict map still denies the un-granted port — least privilege is enforced at **both** the CIDR and the port granularity, and the default at each is **deny** (I6/R3).

8.4 Need-to-know proven

carol-laptop's NetworkMap contains **only conn-A**. It does not receive conn-B's pubkey, endpoint, or even its existence (S3/R2). Carol cannot reconnoiter the office LAN, cannot attempt a WG handshake to conn-B (no pubkey), and any packet

toward 192.168.50.0/24 is dropped by cryptokey routing (no AllowedIPs). Three independent layers — *filtered map*, *cryptokey routing*, *verdict map* — each default-deny, all agree.

9. Edge cases (defined behavior, no guessing)

Per §11.4.6, every boundary has a *defined* zero-trust behavior. These extend [v03-control-plane/svc-policy.md §10] with the zero-trust/default-deny lens.

#	Edge case	Defined behavior
Z1	Newly-enrolled device, no policy grant	Online but isolated : empty AllowedIPs, no verdict entry, in no peer's VisibleTo ([04-security-privacy-pki.md §1.3]). The acceptance criterion "enroll a client; deny an unauthorized host" made concrete.
Z2	Empty ACL (only groups/hosts)	Compiles to all-empty → everyone denied everything (default-deny floor, E1). Valid, not an error.
Z3	Device shares the tenant /48 with another	No reach without a rule — overlay adjacency confers nothing (§1.2 #3).
Z4	Device obtains a peer pubkey out-of-band	Cannot reach it: G4 AllowedIPs + verdict map drop the packet (no compiled grant). Auth ≠ authorization.
Z5	Control plane unreachable from a device	Fail-static to the last filtered map; cannot widen; kill-switch holds the leak window closed (S9).
Z6	Edge boots with no compiled policy	policy: None ⇒ drop everything until a policy lands (R3 fail-closed).
Z7	Policy update fails to compile	Rejected (P4); prior active policy stays live; never allow-all.
Z8	A granted device is revoked	Removed from every VisibleTo + peer dropped + stream closed in <1 s (S5); re-checked at activation (E5).
Z9	A wildcard *: * grant to a non-admin group	Honored — the operator <i>explicitly</i> chose broad reach (E2). Zero-trust is "no <i>implicit</i> trust"; an explicit broad grant is the operator's authorized decision, still compiled per-device.
Z10	Two groups grant overlapping reach to one user	Union — accept rules are additive (no deny to subtract, E6). The broader <i>explicit</i> grant wins; default-deny still governs everything ungranted.
Z11	Stale delta in flight across edges	Fail-static (each edge enforces last-good); honest sub-second window, never fail-open (§7.2).
Z12		Blocking ERR_EXIT_IS_CONNECTOR — a connector cannot be a full-tunnel exit (E7).

#	Edge case	Defined behavior
	exitNodes names a connector	

10. Test & validation mapping (§11.4.169)

Zero-trust is anti-bluff: each property's PASS is captured evidence that an *unauthorized* attempt was actually **denied**, not merely that an authorized one succeeded — a green "allow" test is necessary, never sufficient ([v03-control-plane/svc-policy.md §12 floor], §11.4.5/§11.4.69/§1.1).

Property	Test type	Concrete test point	Captured evidence
Default-deny floor (S1/E1/Z2)	Unit	Compile(emptyACL) → all maps empty; assert no device reaches any other	golden compiled.json byte-compare
New device isolated (Z1)	E2E (§11.4.98)	enroll a client with no grant → assert empty AllowedIPs + cannot reach any LAN host (packet-level)	docs/qa/<run-id>/ transcript + DROP pcap
Least privilege port grant (§8.3)	E2E	carol reaches 10.10.0.0/24:554 and is denied :22 and 192.168.50.0/24	dual allow/deny pcap
Need-to-know filter (S3/R2/§8.4)	Integration	carol's served map contains only conn-A; conn-B absent	served-map byte-compare
Fail-closed compile (P4/Z7)	Integration + meta-test (§1.1)	spec citing an unadvertised CIDR → Update rejects; assert no policies row inserted ; mutate to skip the gate → test FAILs	DB row-count before/after + mutate→FAIL log
Edge fail-closed (R3/Z6)	Integration	start edge with policy: None → every packet dropped	drop-counter trace
Fail-static, not fail-open (§7.2/Z11)	Chaos (§11.4.85)	kill the policy push mid-delta → edges keep enforcing old policy; assert no allow-all window	recovery_trace.log + pcap (no leaked flow)
Server-authoritative	Security	a device requests another device's map	request/response capture

Property	Test type	Concrete test point	Captured evidence
identity (§6.1 #1)		→ served its own slice only	
Auth ≠ authorization (Z4/G4)	Security	send a packet to a known-pubkey peer with no compiled grant → dropped	DROP pcap
Revoked-exclusion (S5/Z8)	Integration	revoke a granted device → removed from VisibleTo + peer dropped, timed <1 s	revoke-to-removed timer
Two-layer enforcement (§3/§8.3)	Meta-test (§1.1)	(a) widen verdict default to ACCEPT → least-privilege port test FAILs; (b) skip AllowedIPs filter → CIDR-deny test FAILs	mutate→FAIL→restore→PASS
Determinism of the grant (P2)	Unit/property	compile same intent twice → byte-identical artifact (no nondeterministic widening)	hash-equality log

Anti-bluff floor (the §0.1 covenant): a zero-trust "PASS" requires captured evidence that **both** an authorized edge succeeded **and** an unauthorized edge was denied — under the same build, same topology. The §1.1 mutations (skip the dry-run gate; widen the verdict default; drop the need-to-know filter; remove the revoked check) are the runtime signatures (§11.4.108) that S1/S3/P4/I6/R3 are **live**, not merely promised.

Sources verified

- [04-security-privacy-pki.md] — §0.1 invariants S1 (zero-trust default-deny), S3 (need-to-know map distribution), S4 (two-channel auth), S5 (<1 s revocation), S9 (kill-switch); §1.2 two authentication channels; §1.3 default-deny realisation (empty AllowedIPs + no verdict entry = isolated); §4.6 revocation pipeline. (Read 2026-06-25.)
- [v03-control-plane/svc-policy.md] — §0.1 invariants P1 (default-deny/need-to-know), P4 (fail-closed), P6 (RLS); §2.1 running ACL example; §3 CompiledPolicy {Version, SpecHash, VisibleTo, AllowedIPs, Verdicts, Via6, ExitNodes} + §3 P3 two-artifact model; §4.1 compile algorithm (default-deny base); §4.3 worked carol-laptop slice; §5 blocking/advisory taxonomy; §6 version lifecycle + instant rollback; §7.2 Peer projection; §10 edge cases (E1/E2/E6/E7) + §10.3 honest convergence-race boundary; §11 convergence SLO; §12 test matrix. (Read 2026-06-25.)
- [v02-data-plane/routing-and-addressing.md] — §0.2 invariants I5/I6 (default-deny), R2 (need-to-know delivery), R3 (verdict map fails closed); §1

RouteEngine (policy: Option<CompiledPolicy>, None ⇒ fail-closed);
§6 ACL→AllowedIPs + verdict-map compiler. (Read 2026-06-25.)

- [threat-model.md] (this volume) — §5.2/§5.4/§5.5 component STRIDE referenced for the authenticate-vs-authorize separation and the over-share threat. (Authored alongside, 2026-06-25.)
- [04_ARCH] HelixVPN-Architecture-Refined.md §3.4 (overlay/policy, AllowedIPs + nftables/eBPF verdict map, ULA /48 + 4via6), §7 (zero-trust, no-logging) — cited via the sibling specs.
- [04_P1] HelixVPN-Phase1-MVP.md §7 (ACL model, two artifacts, fail-closed), §11 DoD (authorized/denied acceptance criterion) — cited via the sibling specs.
- [research-pki_pq_nat], [SYNTHESIS §1/§3/§7] — trust gradient, D4 4via6, need-to-know. **UNVERIFIED** items (measured <1 s convergence p99 under 10k streams; the exact stale-delta window distribution) are flagged inline per §11.4.6 and stated as targets, never results.